Indian Metro Rental Price Estimator

Real listing preparation, baseline comparison, and calibrated uncertainty

Project type	Applied machine learning
Coverage	Bangalore, New Delhi, Mumbai, Pune, and Nagpur
Raw listings	21,691
Prepared by	Harika

1. Executive summary

Two licensed Indian rental datasets contribute 21,691 raw listings. After validity rules and exact signature deduplication, 12,201 rows remain for modelling.

The held-out MAE is INR 11,652, compared with INR 22,122 for a city-median baseline. The model R2 is 0.685.

A separate calibration split produces a plus or minus INR 26,637 interval with 89.7% coverage on the final test set.

2. Problem and intended use

Rental listings mix location, floor area, furnishing, property configuration, and seller information. A city-wide average cannot represent those interactions.

The project tests whether a reproducible model improves on a transparent baseline and whether its uncertainty can be communicated without presenting an asking rent as an exact value.

The output supports learning and listing comparison. It is not a rent valuation, negotiation instruction, legal recommendation, or guarantee of availability.

3. Sources and provenance

India House Rent Prediction contributes listings from Bangalore, New Delhi, Pune, Mumbai, and Nagpur. New Delhi Rental Properties contributes a larger scraped Delhi snapshot.

Both Kaggle pages state an MIT licence. The acquisition script records page URLs, file names, row counts, and filtering results in the source manifest.

The source snapshots were collected at different times and do not represent a probability sample of Indian renters or properties.

Source	Rows before shared cleaning	Licence
India House Rent Prediction	7,691	MIT
New Delhi Rental Properties	14,000	MIT

4. Data preparation

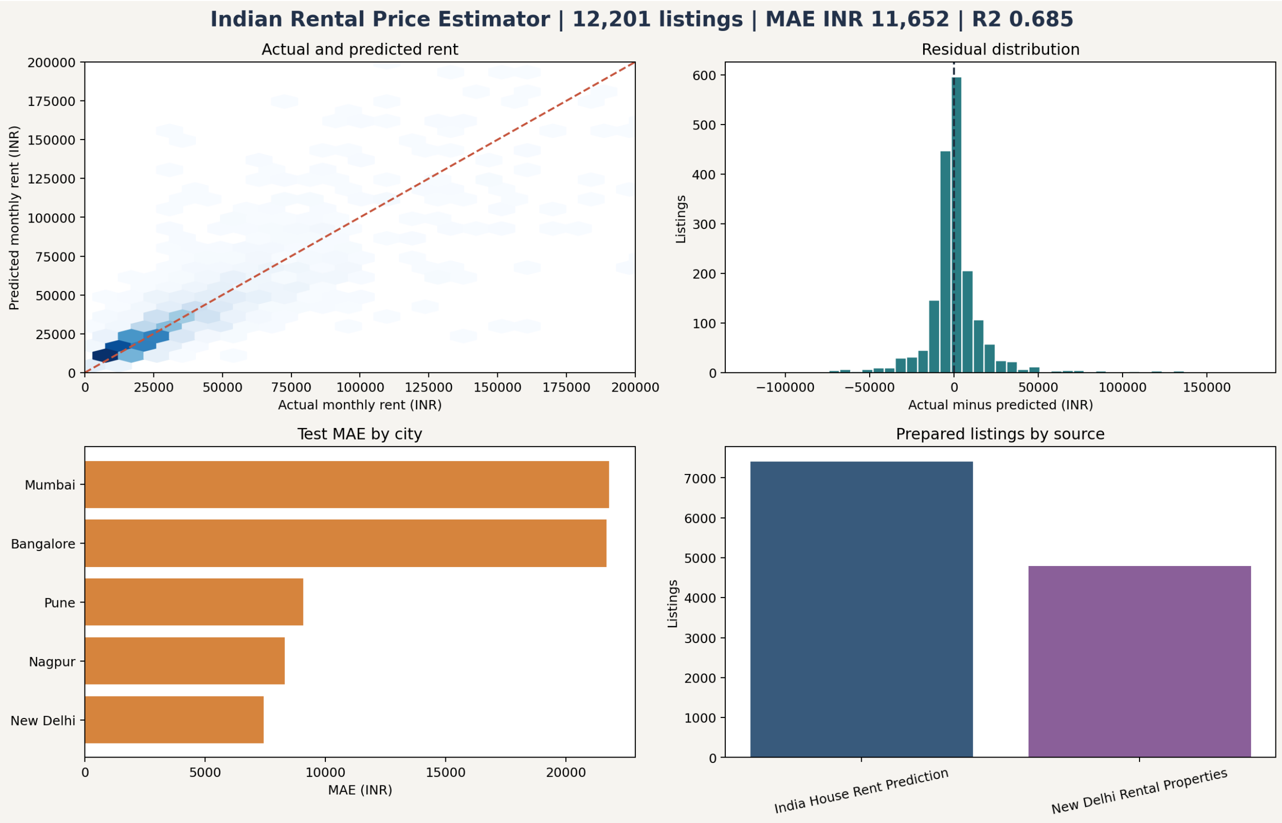
The two schemas are mapped to a common set of twelve fields. 7,798 rows are removed for missing or implausible target, area, or bedroom values. 1,692 exact signatures are removed.

Raw combined data and prepared model data are stored separately. The cleaning rules are testable functions and the final table can be regenerated from the source endpoints.

The prepared table contains city, locality, property type, furnishing, seller type, bedrooms, bathrooms, balconies, area, and monthly asking rent.

Quality rule	Reason
Rent INR 1,000 to 300,000	Remove impossible and extreme entries for this scope
Area 150 to 8,000 sq ft	Remove invalid units and obvious entry errors
Bedrooms 0.5 to 10	Retain RK-style values and remove invalid values
Exact listing signature	Reduce repeated listings without fuzzy deletion

5. Evaluation dashboard



The evaluation page combines prediction fit, residual shape, city-level MAE, and prepared source mix so that one headline metric does not hide important variation.

6. Model and baseline

Categorical fields use an ordinal encoder that preserves an explicit unknown category. Numeric gaps use median imputation. Histogram gradient boosting learns nonlinear effects without requiring a very large one-hot matrix.

The comparison baseline predicts the median training rent for each city. This is simple, reproducible, and difficult to beat by chance when city is a dominant factor.

The model improves MAE by 47.3% over that baseline on the untouched test set.

Metric	Result
MAE	INR 11,652
Baseline MAE	INR 22,122
RMSE	INR 22,696
R2	0.685

7. Validation and uncertainty

The split uses 8,540 training rows, 1,830 calibration rows, and 1,831 final test rows, stratified by city.

Absolute calibration residuals define a 90th-percentile interval radius. This split-conformal style construction does not assume normally distributed errors.

Final interval coverage is 89.7%. The interval is intentionally wide because listing noise and omitted variables remain substantial.

8. Error analysis

Bangalore and Mumbai have higher MAE than New Delhi, Pune, and Nagpur in this prepared sample. Those city results are published in the evaluation JSON instead of being averaged away.

Higher-price listings contribute disproportionately to RMSE. Residual review also shows that omitted building condition, maintenance, exact position, and timing remain important.

A next version should test time-aware validation when trustworthy listing dates are available and should publish error by rent band and locality support.

9. Limitations, reproducibility, and next steps

Asking rent is not contracted rent. Scraped listings may be stale, promotional, repeated across sites, or entered incorrectly. The data does not cover every Indian city or locality evenly.

The repository contains acquisition, transformation, modelling, evaluation, prediction, tests, visual evidence, and this report. The README lists one command sequence for a complete rebuild.

Next steps are to add a linear baseline, obtain reliable collection dates, calibrate city-specific intervals, review locality support, and create a small interface that always shows the model date and uncertainty.